

LaTeX for Research Writing

AIHA8121: Research Methodology & Ethics

Week	Content Coverage	Date	
1	Introduction to Research Methodology and Ethics	2026/9/5	2026/9/5
2	Literature Review	2026/9/12	2026/9/18下午
3	Learning LaTeX & Writing Research Proposals I	2026/9/19	2026/9/19
4	Writing Research Proposals II	2026/9/26	2026/10/9下午
5	Writing Research Papers I	2026/10/3	2026/10/10
6	Writing Research Papers II	2026/10/10	2026/10/16下午
7	Writing Research Papers III	2026/10/17	2026/10/17
8	Research Ethics	2026/10/24	2026/10/24
9	Making Posters and Slides	2026/10/31	2026/10/31
10	Making Posters and Slides	2026/11/7	2026/11/7
11	Making Posters and Slides	2026/11/14	2026/11/14
12	Individual Student Presentation I	2026/11/21	2026/11/21
13	Individual Student Presentation II	2026/11/28	2026/11/28
14	Individual student presentation III	2026/12/5	2026/12/5
15	Individual student presentation IV	2026/12/12	2026/12/12

CONTENTS

01 Workflow and environment

02 Build a research paper

03 Debug, deliver, and extend

SECTION 01

Workflow and Environment

Understand the software layers before learning individual commands

`SOURCE FILES: main.tex · references.bib · optional figures/`

What is LaTeX?



- LaTeX is pronounced “lay-tech” or “lah-tech,” not “la-teks.”
- LaTeX is a document preparation system for high-quality typesetting.
- It is not a word processor, but is used as a document markup language.
- LaTeX is most often used to produce technical or scientific documents, but it can be used for almost any form of publishing.
- It was originally written by Leslie Lamport and is based

on the TeX typesetting engine by Donald Knuth.

Why Use LaTeX?



- Designed by academics and easily accommodates academic use.
- Professionally crafted predefined layouts make a document really look as if “printed.”
- Mathematical symbols and equations are easily integrated.
- Even complex structures such as footnotes, references, table of contents, and bibliographies can be generated easily.
- Forces author to focus on logical instead of aesthetic structure of a document.
- Creates more beautiful documents.
- Portable, compatible, flexible, versatile, and cheap (or free)!
  - It is available for Windows, Linux, Mac OS X, and others
  - Its file format is plain text – readable and editable.
  - It will produce the same output on all operating systems.
  - Produces the output in PostScript, PDF, DVI, and HTML format.

Where LaTeX Helps



Researchers use LaTeX for

- equations and technical notation
- automatic numbering and cross-references
- citations and bibliographies
- theses, papers, and other long documents
- plain-text source and version history

Consider another tool

- rapid free-form page design
- short documents with heavy direct editing
- teams that require tracked visual edits in one file
- workflows where a mandated template uses another format

Trade-off: compiling has a learning curve and requires reading error messages.

LaTeX is most valuable when structure, numbering, and long-term maintenance matter

How LaTeX works?

SOURCE	PROCESS	OUTPUT
Plain text and structural commands in a .tex file	**A TeX engine interprets the source**	**A Consistently Typeset PDF** Consistent type, numbering, equations, and references.

LaTeX separates what the content means from how the page is typeset.

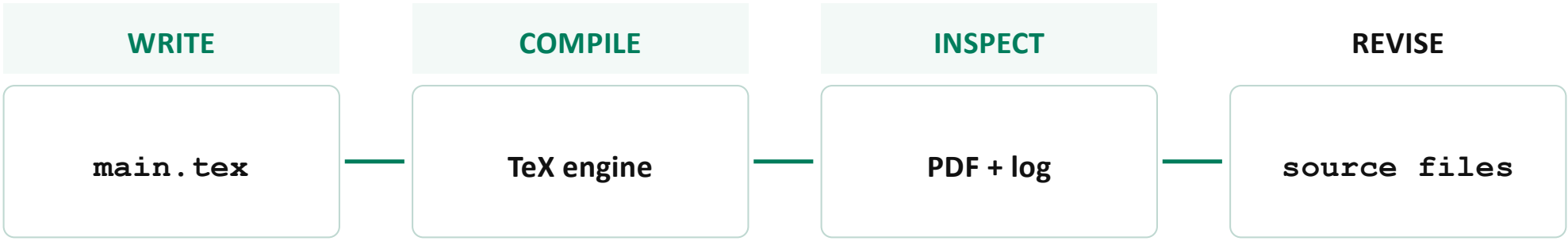
The source describes document structure; the compiler generates the page

The LaTeX Ecosystem

PROJECT FILES	`main.tex` `references.bib` `figures/ (optional)`
EDITOR OR PLATFORM	Overleaf · TeXstudio · VS Code with a LaTeX extension
TeX DISTRIBUTION	TeX Live · MacTeX · MiKTeX
ENGINE	pdfLaTeX · XeLaTeX · LuaLaTeX

A platform, an editor, a distribution, and a compiler are different layers

From Source to PDF



references.bib and figures/ enter the build when the source calls them

Some workspaces automate repeated build steps; the log still records them.

Writing in LaTeX is a repeated source–compile–inspect–revise loop

Choosing a Compiler

pdfLaTeX	XeLaTeX	LuaLaTeX
Reliable default for many English journal templates	**Unicode and direct access to modern system fonts**	**Unicode, modern fonts, and Lua-based extensions**
PNG · JPG · PDF	Strong choice for CJK	Useful for advanced workflows

Workshop default: pdfLaTeX for this English example. Use XeLaTeX or LuaLaTeX for multilingual text when the template allows it.

Follow the template first; choose another engine for a specific language or font need

Online or Local?



Online workspace

- editor, compiler, and PDF preview in one browser window
- easy sharing for class work and collaboration
- no local installation for a first lesson

Local setup

- an editor plus a TeX distribution
- useful for offline, private, or large projects
- versions and fonts can be controlled across collaborators

HANDS-ON： SETUP

Local LaTeX Setup

SYSTEM	TeX DISTRIBUTION	EDITOR
Windows	**TeX Live or MiKTeX**	TeXstudio or VS Code with a LaTeX extension
macOS	**MacTeX**	TeXShop, TeXstudio, or VS Code with a LaTeX extension
Linux	**TeX Live from a supported source**	TeXstudio, VS Code with a LaTeX extension, or another editor

Verify the installation with the minimal document before opening a complex template.

Install a supported distribution, choose an editor, then test the smallest document

Installing LaTeX



- In Windows
- MiKTeX
  - MiKTeX is a typesetting system for the Windows.
  - Download from www.miktex.org for free
  - It is generally recommended to install MiKTeX first, then WinEdt.
- WinEdt
  - WinEdt is a text editor.
  - WinEdt creates the source file (.tex and others).
  - Download from www.winedt.com for free for 30 days.
  - WinEdt costs $30.

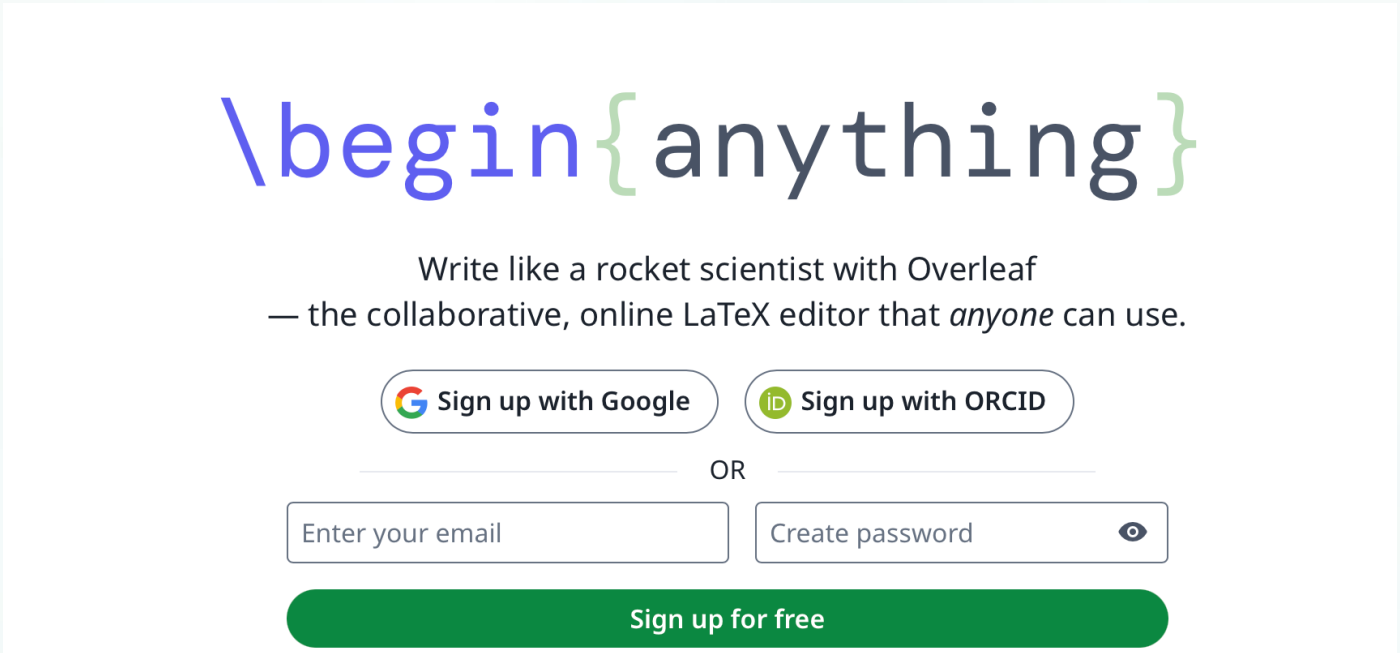
Installing LaTeX



- Other text editors
  - There are other text editors.
  - Winshell for free (http://www.winshell.de/)
  - Scientific Workplace
    - Combination of LaTeX and Mathematics program
    - Does a good job of calculating and graphing, very user friendly, but expensive

- In Mac
  - TexShop
  - Download for free http://www.uoregon.edu/~koch/texshop/
  - Includes everything!

https://www.overleaf.com



SECTION 02A



Document Structure and Text

Start with a compiling baseline, then add one structural element at a time

FOUR WORKING LINES · ONE NEW ELEMENT · COMPILE AGAIN

A Minimal Document

SOURCE

```
\documentclass[12pt]{article}
\begin{document}
Hello, AIHA8121.
\end{document}
```

COMPILED OUTPUT

Hello, AIHA8121.

HANDS-ON：Compile these four lines in main.tex, replace the sentence with your name, and compile again.

A reliable baseline needs a document class and a complete document environment

Preamble and Document Body



FULL FILE AFTER THIS STEP

```
1  \documentclass[12pt]{article}
2  \usepackage{amsmath}
3  \usepackage{graphicx}
4  % global settings
5
6  \begin{document}
7  Hello, Your Name.
8  \end{document}
```

PREAMBLE

- **Document class**
- **Packages**
- **Global settings**

DOCUMENT BODY

- **Title and abstract**
- **Sections and evidence**
- **Bibliography output**

The preamble enables document-wide behavior; the body contains what readers see

Document Classes and Packages



Document class selects the document family

`article`	papers and short reports
`report`	longer reports with chapters
`beamer`	presentation slides

Packages add specific capabilities

ADD ONLY THE % ADD NOW LINES

```
% ALREADY IN YOUR FILE
\documentclass[12pt]{article}
\usepackage{amsmath}
\usepackage{graphicx}

% ADD NOW
\usepackage{booktabs}
\usepackage[backend=biber]{biblatex}
\addbibresource{references.bib}
```

Load only what the paper uses. A journal template may already make these choices.

The class defines the document family; packages add named capabilities

Title and Abstract

```
% PREAMBLE, before \begin{document}
\title{Illustrative Sensor-Based Prediction of
Six-Month Fall Risk}
\author{Your Name}
\date{}

% DOCUMENT BODY, after the existing
\begin{document}
\maketitle
\begin{abstract}
A one-page smart-ageing typesetting example.
All people, scores, and results are
illustrative.
\end{abstract}
```

Illustrative Sensor-Based Prediction of Six-Month Fall Risk

Your Name

Abstract

A one-page smart-ageing typesetting example. All people, scores, and results are illustrative.

Define metadata in the preamble; place \maketitle in the document body.

The front matter identifies the paper and states the status of its evidence

Sections, Lists, and Paragraphs

ADD INSIDE THE DOCUMENT BODY

```
 1  \section{Model}
 2  Six-month fall risk is modeled from
a sensor-derived score.
 3
 4  \begin{itemize}
 5    \item Sensor risk score \(s\)
 6    \item Predicted fall probability
\(p(s)\)
 7  \end{itemize}
 8
 9  This blank source line starts a new
paragraph.
```

1 Model

Six-month fall risk is modeled from a sensor-derived score.

- Sensor risk score s
- Predicted fall probability p(s)

This blank source line starts a new paragraph.

A single source line break acts like a space. A blank line starts a paragraph.

Use section and list environments for structure; let LaTeX maintain numbering and spacing

Spacing and Special Characters



```
 1  % This comment stays in the source.
 2  Multiple     spaces appear as one
space.
 3
 4  A blank line starts a new paragraph.
 5
 6  Eighty percent is written as 80\%.
 7  File names may contain fall\_risk.
 8  Reserved characters: \# \$ \% \& \_ \{
\}
 9  Backslash: \textbackslash
```

Multiple spaces appear as one space.

A blank line starts a new paragraph.

Eighty percent is written as 80%.

File names may contain fall_risk.

Reserved characters: # $ % & _ { }

Backslash: \

Whitespace is structural, comments stay in the source, and reserved characters must be escaped

Emphasis and Alignment

```
 1  The values are \emph{illustrative}.
 2  This example is \textbf{not
empirical}.
 3
 4  \begin{center}
 5  All numerical values are
illustrative.
 6  \end{center}
 7
 8  \begin{figure}[htbp]
 9    \centering
10    % figure content
11  \end{figure}
```

The values are *illustrative*.

This example is **not empirical**.

All numerical values are illustrative.

Use \centering inside a figure.
It is a declaration; center is an environment.

Use semantic emphasis and choose center or centering according to scope

Labels and Cross-References



EDIT THE EXISTING SECTION LINE

```
\section{Model}\label{sec:model}
See Section~\ref{sec:model}.
```

OBJECT	LABEL
Section	sec:model

See Section 1.

If you see ??, compile again.
Then recheck the label key.

Label the object, refer to the label, and let LaTeX maintain the number

SECTION 02B

Equations, Figures, Tables, and References

Research objects become useful when the text introduces, labels, and interprets them

FORMULA · FIGURE · TABLE · REFERENCE

Inline and Displayed Mathematics

```
An illustrative prediction is
\(p_i=0.31\).

\[
p(s)=\frac{1}{1+\exp[-(\beta_0+\beta_1
s)]}
\]
```

COMPILED OUTPUT

Inline: $p_i = 0.31$

Display:
$$p(s) = \frac{1}{1 + \exp[-(\beta_0 + \beta_1 s)]}$$

s sensor-derived risk score

p(s) predicted six-month fall probability

β_0 intercept

β_1 change in log odds per score unit

Use inline mathematics inside a sentence and display mathematics for an expression that needs focus

Numbered and Aligned Equations



```
\begin{align}
\eta_i &= \beta_0+\beta_1 s_i
  \label{eq:score}\\
p_i &= \frac{1}{1+\exp(-\eta_i)}.
\end{align}

Equation~\eqref{eq:score} defines the linear
predictor.
```

$$\eta_i = \beta_0 + \beta_1 s_i \qquad (1)$$

$$p_i = \frac{1}{1 + \exp(-\eta_i)} \qquad (2)$$

```
& sets alignment
\\ ends a line
\label connects to \eqref
```

Use equation for one numbered expression and align for related lines with a shared alignment point

Figures with Captions and Labels

```
 1  \begin{figure}[htbp]
 2    \centering
 3    \fbox{\parbox[c][1.2in][c]
 4      {.68\linewidth}{
 5        \centering Illustrative fall-
risk curve}}
 6    \caption{Illustrative six-month
fall-risk
 7      curve placeholder.}
 8    \label{fig:risk}
 9  \end{figure}
10  Figure~\ref{fig:risk} shows the
placeholder.
```

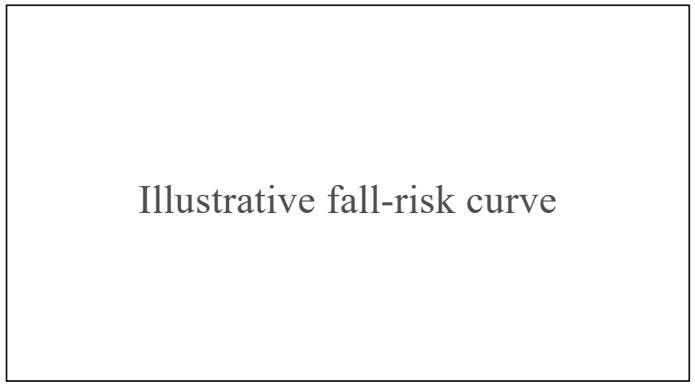


Figure 1. Illustrative six-month fall-risk curve placeholder.

No external image is required.

[htbp] suggests positions; it does not force placement.

A figure needs a caption, a label, and a sentence that refers to it

Tables with Booktabs

```
\begin{table}[htbp]
\centering
\caption{Illustrative sensor scores and
predicted fall risks.}
\label{tab:data}
\begin{tabular}{rr}
\toprule
Sensor risk score & Predicted risk \\
\midrule
0.0 & 0.12 \\
1.0 & 0.27 \\
2.0 & 0.50 \\
\bottomrule
\end{tabular}
\end{table}
```

Table 1. Illustrative sensor scores and predicted fall risks.

Sensor risk score	Predicted risk
0.0	0.12
1.0	0.27
2.0	0.50

`&` separates cells `\\` ends a row

The table float owns the caption and number; tabular owns the rows and columns

Citations and Bibliography

references.bib

```
@article{howcroft2013,
  author  = {Howcroft, Jennifer and Kofman,
Jonathan
             and Lemaire, Edward D.},
  title   = {Review of fall risk assessment
in geriatric
             populations using inertial
sensors},
  journal = {Journal of NeuroEngineering
and Rehabilitation},
  year    = {2013},
  doi     = {10.1186/1743-0003-10-91}
}
```

main.tex

```
Inertial sensors have been reviewed for
fall-risk
assessment in older adults
\cite{howcroft2013}.
\printbibliography
```

Inertial sensors have been reviewed for fall-risk assessment in older adults [1].

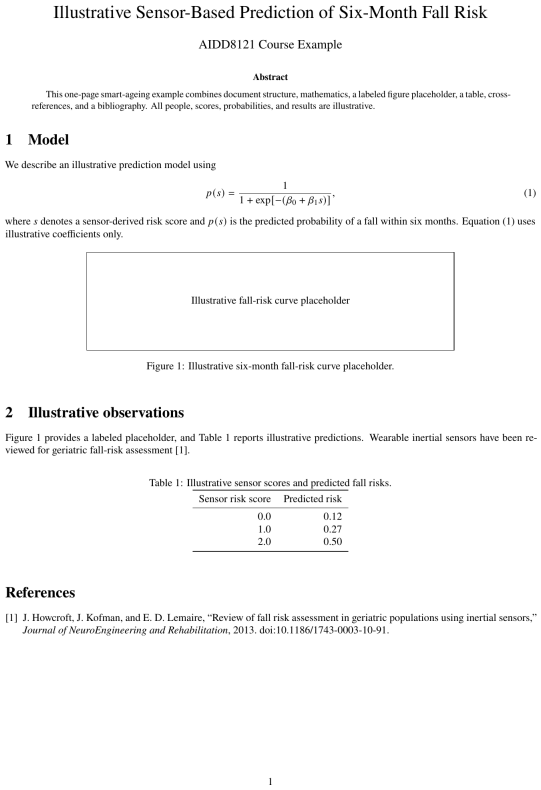
References

[1] J. Howcroft, J. Kofman, and E. D. Lemaire. Review of fall risk assessment in geriatric populations using inertial sensors. Journal of NeuroEngineering and Rehabilitation, 2013.

No Biber? Compare source and output in observation mode.

A citation key connects a verified claim to structured metadata and a formatted reference

One Project Becomes a Short Paper



Illustrative Sensor-Based Prediction of Six-Month Fall Risk

AIDD8121 Course Example

Abstract

This one-page smart-ageing example combines document structure, mathematics, a labeled figure placeholder, a table, cross-references, and a bibliography. All people, scores, probabilities, and results are illustrative.

1 Model

We describe an illustrative prediction model using

$$p(s) = \frac{1}{1 + \exp[-(\beta_0 + \beta_1 s)]}, \tag{1}$$

where s denotes a sensor-derived risk score and $p(s)$ is the predicted probability of a fall within six months. Equation (1) uses illustrative coefficients only.

Illustrative fall-risk curve placeholder

Figure 1: Illustrative six-month fall-risk curve placeholder.

2 Illustrative observations

Figure 1 provides a labeled placeholder, and Table 1 reports illustrative predictions. Wearable inertial sensors have been reviewed for geriatric fall-risk assessment [1].

Table 1: Illustrative sensor scores and predicted fall risks.

Sensor risk score	Predicted risk
0.0	0.12
1.0	0.27
2.0	0.50

References

[1] J. Howcroft, J. Kofman, and E. D. Lemaire, "Review of fall risk assessment in geriatric populations using inertial sensors," *Journal of NeuroEngineering and Rehabilitation*, 2013. doi:10.1186/1743-0003-10-91.

1

The compiled page now contains

01	**Structure**	title, abstract, and numbered sections
02	**Model**	a typeset and referenced fall-risk equation
03	**Evidence**	a labeled risk-curve placeholder and a table
04	**Traceability**	cross-references and a bibliography
05	**Boundary**	all numerical values marked illustrative

Every numbered object is introduced and interpreted in the surrounding text.

SECTION 03



Debugging and Delivery

A clean PDF begins with a disciplined reading of the build log

`FIRST ERROR · SOURCE CONTEXT · ONE REPAIR · COMPILE AGAIN`

Read the First Real Error

```
! LaTeX Error: \begin{abstract}
  on input line 12 ended by
\end{document}.
l.27 \end{document}
```

Diagnosis
The abstract environment is missing \end{abstract}.

1. **Read the first error**
   Later messages may be consequences.
2. **Open nearby source**
   Read context, not only the line number.
3. **Check structure**
   Check braces, environments, math, packages, and paths.
4. **Repair one cause**
   Compile again; confirm the first error changes.

One early structural error can trigger many later messages, so repair the first cause first

Debugging Activity

PAIR WORK · 8 MINUTES

```
 1  \section{Results
 2  \textbff{Illustrative results}
 3  The prediction variable is
fall_risk_score.
 4  \begin{tabular}{cc}
 5  A & B \\
 6  1 & 2 & 3 \\
 7  \end{tabular}
```

Find and classify four problems

STRUCTURE	an unclosed argument
COMMAND	an undefined control sequence
TEXT	an unescaped underscore
TABLE	too many cells in the second row

Fix one issue at a time; record the first log message after each compile.

Classify the first error before editing; the error category suggests the next check

Common Mistakes



- Often times, you make a mistake when creating a document. You will notice the log file reporting a problem.
- There are some common mistakes:
  - “end” doesn’t follow “begin”
  - $ doesn’t follow $
  - Using commands from packages lot defined in the preamble
  - Don’t forget “\”s.

A Traceable Writing Loop

01

WRITE

Structure and evidence in source files

02

COMPILE

Build with a defined toolchain

03

CHECK

Inspect the PDF, log, labels, and citations

04

DELIVER

Package the files required by your course or venue

Before submission

- every figure and table is cited
- references and labels resolve
- evidence status and limitations are accurate

LaTeX automates formatting and numbering. The researcher remains responsible for evidence and interpretation.

Structured source, a defined build, and honest evidence boundaries make a document easier to rebuild and audit

Appendix · Multilingual Text

REFERENCE ONLY · Use this when the required compiler and ctex are available.

English-only journal template

```
% Follow the journal template.
% pdfLaTeX is often supported.
\documentclass{article}
```

Chinese or multilingual document

```
% Compile with XeLaTeX or LuaLaTeX.
\documentclass{ctexart}

Research confidence (研究信心)
```

Keep the compiler, TeX Live version, packages, and fonts consistent across collaborators.

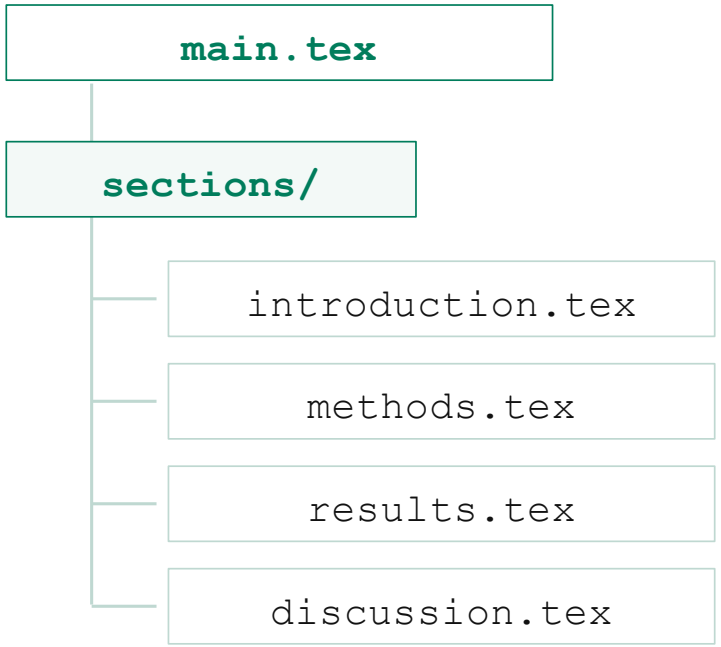
Do not add inputenc or fontenc to a XeLaTeX/ctex example unless a specific template requires it.

Choose one multilingual toolchain and keep it stable throughout the project

Appendix · Multi-File Projects

REFERENCE PATTERN · These section files are not part of today's one-page workshop.

```
1  % main.tex
2  \tableofcontents
3
4  \input{sections/introduction}
5  \input{sections/methods}
6  \input{sections/results}
7  \input{sections/discussion}
```



Only main.tex owns the preamble and document environment.

One main file controls the build; section files keep long projects manageable

Appendix · Multiple Columns

```
% PREAMBLE
\usepackage{multicol}

% DOCUMENT BODY
\begin{multicols}{2}
Short content begins in the first
column.

More content flows into the next column.
\end{multicols}
```

A short two-column example

Short content begins in the first column.

More content flows into the next column.

A journal's two-column class is different from adding a local multicol environment.

Use columns because the template or document purpose requires them, not as decoration

Appendix · Custom Commands

```
% PREAMBLE
\newcommand{\ICfifty}{\mathrm{IC}_{50}}
\newcommand{\parameter}[1]{%
  \ensuremath{#1}}

% DOCUMENT BODY
The illustrative \ICfifty\ is reported in
\(\mu\mathrm{M}\).

\parameter{R_{\max}} \quad \parameter{R_{\min}}
```

The illustrative IC_{50} is reported in µM.

Rmax Rmin

COMMAND NAME	`\parameter`
ARGUMENTS	`1`
PLACEHOLDER	`#1`

A custom command centralizes repeated notation so one definition controls every later use

Custom Commands

```
\newcommand{\TUG}{TeX Users Group}
\begin{document}
\section{Creating Custom Commands}
The \TUG\ is an organization for people who are
interested in \TeX\ or \LaTeX.
\end{document}
```

1 Creating Custom Commands

The TeX Users Group is an organization for people who are interested in TEX or LATEX.

Creating a Macro

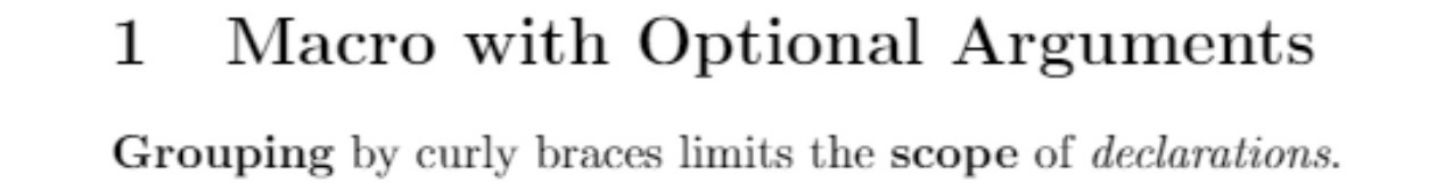
```
\documentclass{article}
\newcommand{\keyword}[1]{\textbf{#1}}
\begin{document}
\section{Defining a Macro and using it}
\keyword{Grouping} by curly braces limits the
\keyword{scope} of \keyword{declarations}.
\end{document}
```

1 Defining a Macro and using it

Grouping by curly braces limits the **scope** of **declarations**.

Command with Optional Arguments

```
\documentclass{article}
\newcommand{\keyword}[2][\bfseries]{{#1#2}}
\begin{document}
\section{Macro with Optional Arguments}
\keyword{Grouping} by curly braces limits the
\keyword{scope} of
\keyword[\itshape]{declarations}.
\end{document}
```



`\newcommand{command}[arguments][optional]{definition}`

`command`	The name of the new command, starting with a backslash followed by lowercase and/or uppercase letters or a backslash followed by a single non-letter symbol. That name must not be already defined and is not allowed to begin with `\end`.
`arguments`	An integer from 1 to 9, the number of arguments of the new command. If omitted, the command will have no arguments.
`optional`	If this is present, then the first of the arguments would be optional with a default value given here. Otherwise all arguments are mandatory.
`definition`	Every occurrence of the `command` will be replaced by `definition` and every occurrence of the form `#n` will then be replaced by the nth argument.

Have you finished…

- Install
- Template
- Help
- Equation: Different types of Eqns. and symbols; alignment
- Figure: image size and rotation
- Table: three-line table
- Reference: hyperlink



https://www.latex-project.org/help/documentation/
https://www.learnlatex.org/
https://www.overleaf.com/learn/latex/Tutorials
https://ctan.org/

Thank You!

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